

Which neural network architecture to use for each problem type

Task		Input		Architecture	Key Reason	Example
Image	Classifi- cation	Image		CNN	Spatial locality, weight sharing; filters: edges $\rightarrow$ shapes $\rightarrow$ objects	ResNet, Effi- cientNet
Object	Detec- tion	Image		CNN + Region Head	Multi-scale feature maps; backbone CNN + detec- tion head	YOLO, Faster R- CNN
Named	Entity	Token	se-	BiLSTM-CRF / BERT	Bidirectional context per token; CRF models label transitions	spaCy, BERT-NER
Text	Classifica- tion	Text	se-	BERT / CNN / LSTM	Global context or local n -grams; BERT $\gg$ LSTM $\gg$ CNN accuracy	BERT fine- tune
Seq-to-Seq		Source	se-	Transformer Enc-Dec	Attention over full source; Encoder reads; decoder attends	T5, BART, mT5
Language	Mod- elling	Token	se-	Transformer De- coder (GPT)	Causal left-to-right at- tention; Masked self- attn: no future tokens	GPT-2/3/4, LLaMA
Time Series		Numeric	se-	LSTM / Trans- former	Temporal state or atten- tion; LSTM moderate; Trans. long-range	Temporal Fusion Trans.
Question	An- swering	[Question, Context]		BERT + Span Head	Bidirectional under- standing; Predicts start/end span in con- text	BERT-large (SQuAD)
Semantic Search		Text		Bi-encoder (SBERT)	Dense vector similarity; Encode query & docs; co- sine sim.	FAISS, Pinecone

Rule of Thumb

Image $\rightarrow$ CNN. **Text** / **sequence** $\rightarrow$ BERT or Transformer (LSTM if low resources).
Token-level $\rightarrow$ BiLSTM-CRF / BERT token head. **Generation** $\rightarrow$ Transformer decoder (GPT-style).